

alliGATOR

Graph Anomaly Detection through Diffusion based Topology Reconstruction

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Problem Background

- People spend increasing amount of time online
 - Sharing personal information directly or not
- 150% year-on-year increase of identity fraud across Europe
- Identity theft and fraudulent transactions cost Europe
€6.4 billion a year

Problem Background

- Detecting criminal activity
 - Monitoring systems analyse monetary transactions
 - Flag suspicious behaviour
- Monetary transactions — graph-structured data
 - Nodes are bank account holders
 - Edges are transactions

Problem Background

- Detecting criminal activity
 - Monitoring systems analyse monetary transactions
 - **Flag suspicious behaviour**
- Monetary transactions — graph-structured data
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Graph Anomaly

Node anomaly

- Out of ordinary activity pattern for its attributes
- Example : Identity theft

Edge anomaly

- Connection between nodes that do not typically interact
- Example : Fraudulent transaction

Approaches to Anomaly Detection

Approaches to Anomaly Detection

Supervised and weakly-supervised approaches

- Require labelled data

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Unsupervised approaches

- Isolation Forest and k-nearest neighbours

Approaches to Anomaly Detection

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Using generative models

- **Idea** : generative models learn to generate normal data, therefore, any deviations from what they have generated can be considered anomalous

Graph Generative Models

Classical models

- Erdős-Rényi, configuration, Barabási-Albert

Machine learning models

- Variational auto-encoders, generative adversarial networks, diffusion models

Diffusion models

- Address instability during training and mode collapse of GANs, limited expressiveness of VAE
- Node attributes, edge structure, both
- Continuous or discrete approach to generation

Anomaly Detection using Graph Generative Models

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Previously presented models are designed to generate realistic synthetic graphs

- Model the global, macro-level properties of graphs

Anomaly Detection using Graph Generative Models

Previously presented models are designed to generate realistic synthetic graphs

- Model the global, macro-level properties of graphs

To detect anomalies on node and edge level we are interested in local, micro-level properties

- Contextualise the generation process by conditioning it on one of the graph components

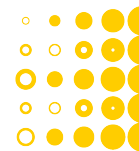
Anomaly Detection using Graph Generative Models (Assumption)

Conditioning the generation on the nodes attributes

- Generate the edges (adjacency matrix, topology)
- Node connections are inherent to their attributes

Conditioning the generation on the graph topology

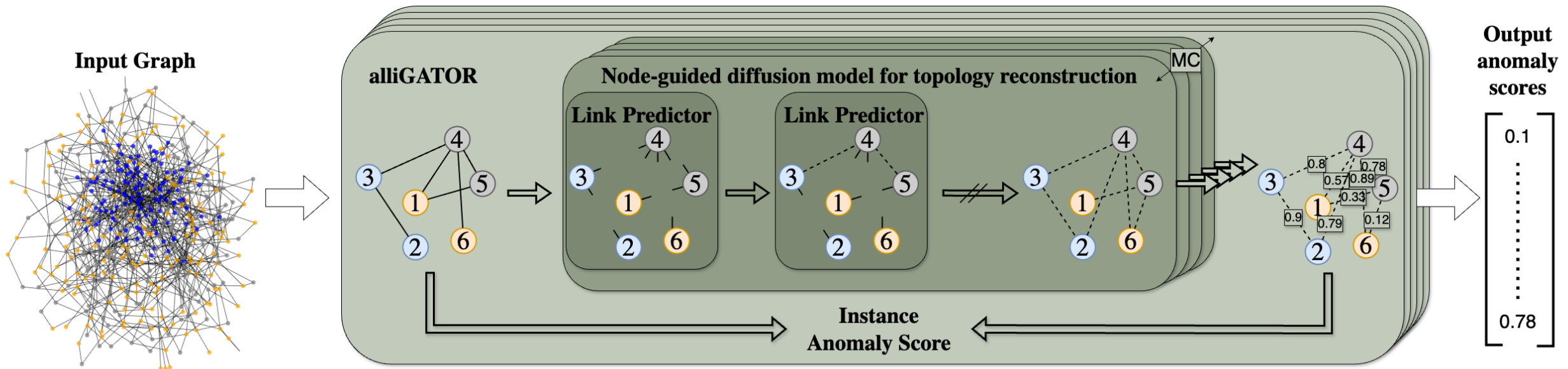
- Generate the node attributes
- Node attributes are dependent on their connections
- Li et al (2023) presented anti-money laundering model



Research Question

Can graph topology reconstruction of ego-networks be used to detect anomalous nodes or edges?

The alliGATOR model: Graph Anomaly Detection through diffusion based Topology Reconstruction



Contributions

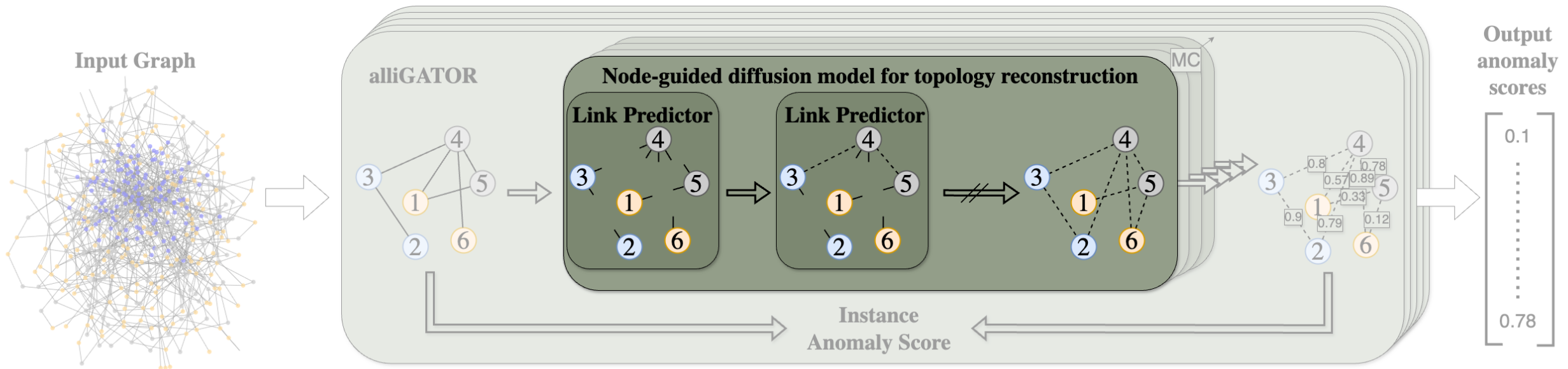
The alliGATOR model

- Unsupervised model that detects anomalies in graph-structured data by modelling graph local-properties

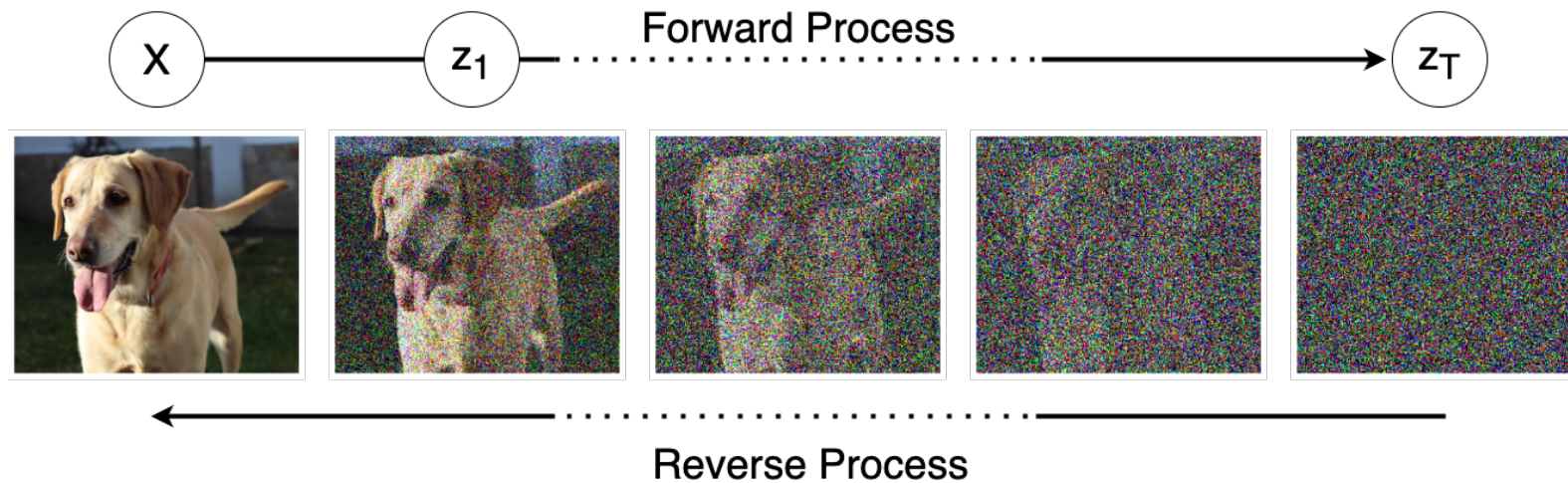
Node-guided diffusion model for topology reconstruction

- Adapted from EDGE model by Chen et al. (2023)
 - Graph generator focused on the adjacency matrix
 - Scalability to thousands of nodes

The alliGATOR model: Graph Anomaly Detection through diffusion based Topology Reconstruction



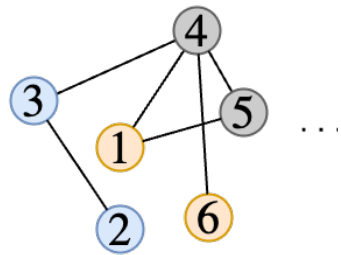
Diffusion Models



Put simply, a diffusion model learns to reverse the incremental addition of noise to data

Forward Process

- *sampling of adjacency matrices (denoted by A)*

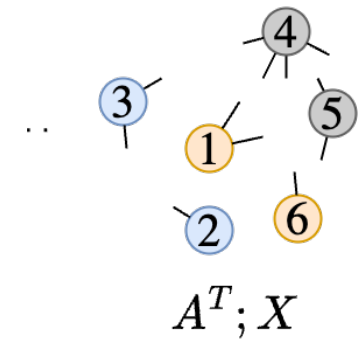
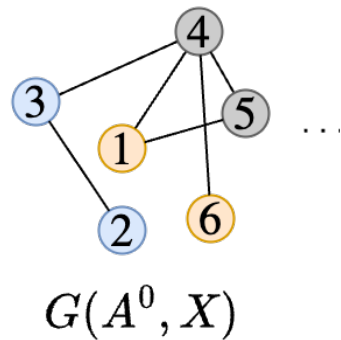


$G(A^0, X)$

Forward Process

Forward Process

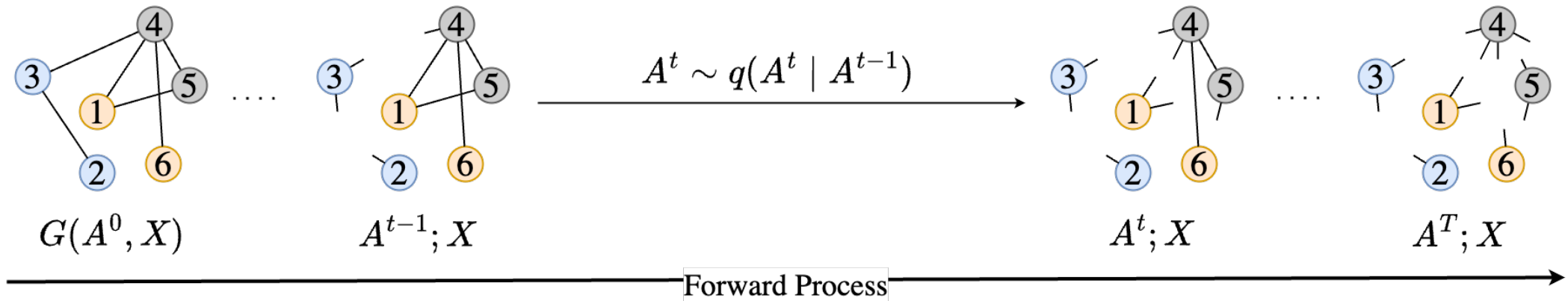
- *sampling of adjacency matrices (denoted by A)*



Forward Process

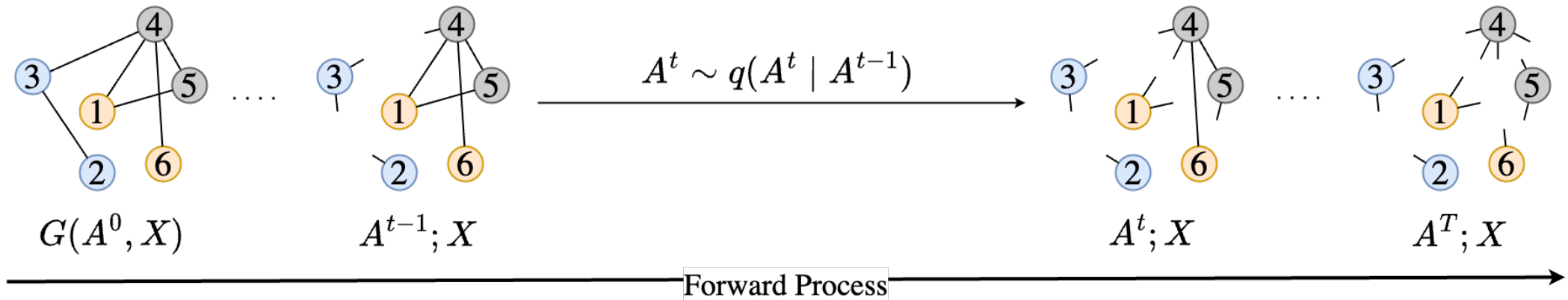
Forward Process

- *sampling of adjacency matrices (denoted by A)*



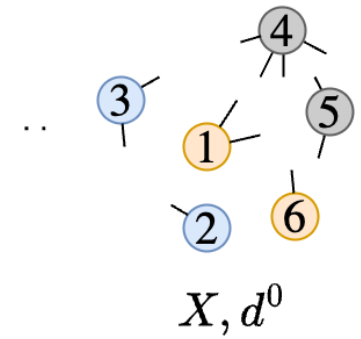
Forward Process

- *sampling of adjacency matrices (denoted by A)*



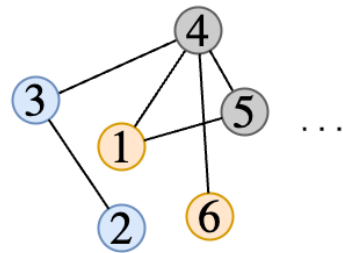
$$\mathcal{B}(\mathbf{A}^t; (1 - \beta_t)\mathbf{A}^{t-1} + \beta_t p)$$

Reverse Process

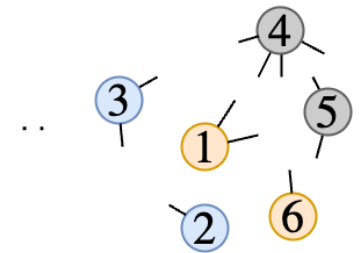


Reverse Process
(Topology Reconstruction)

Reverse Process



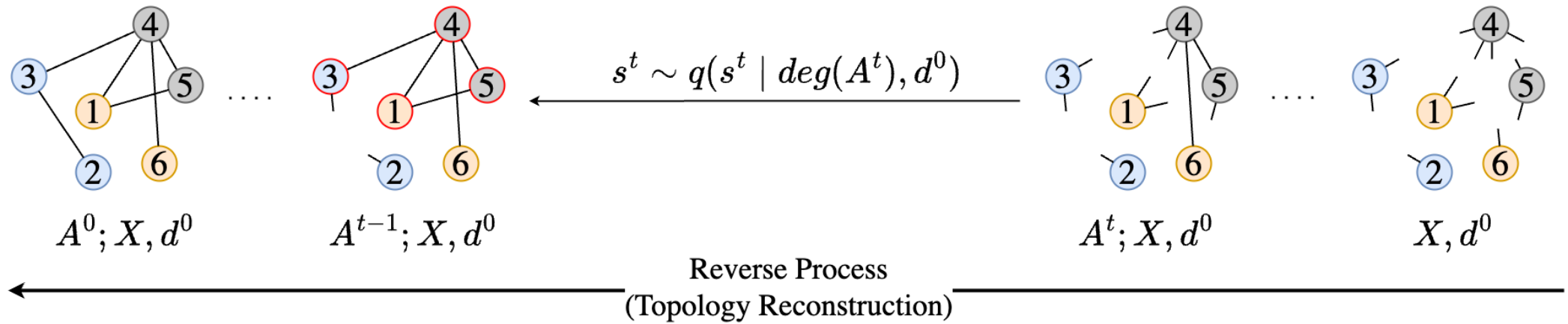
$A^0; X, d^0$



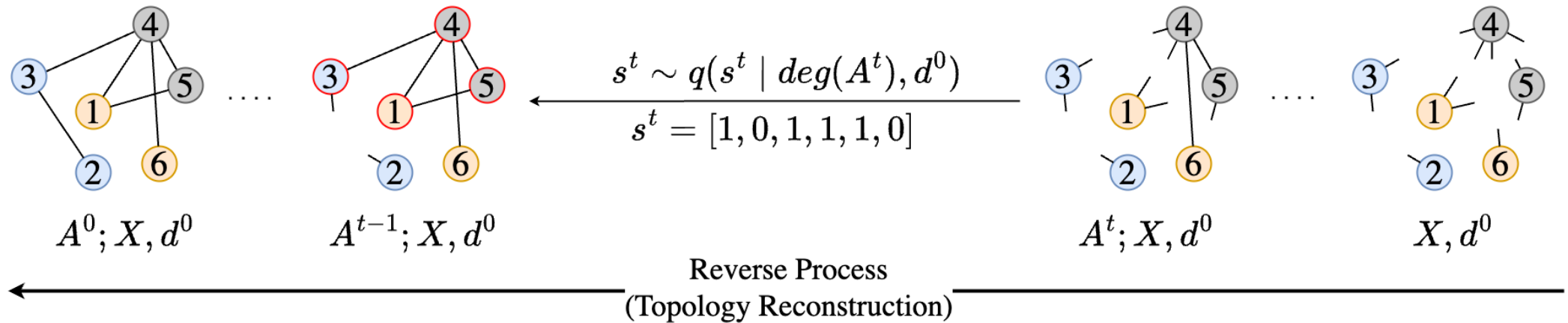
X, d^0

Reverse Process
(Topology Reconstruction)

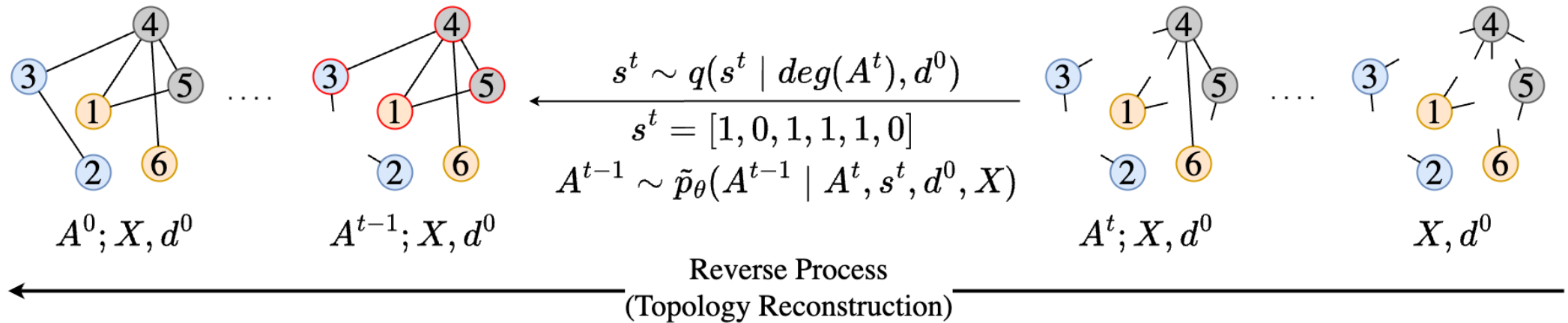
Reverse Process



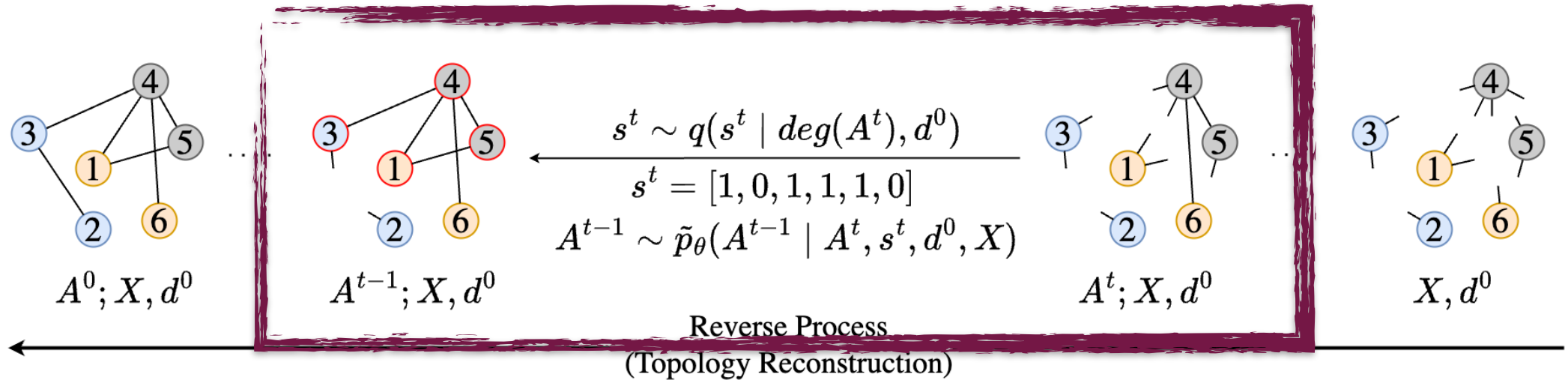
Reverse Process



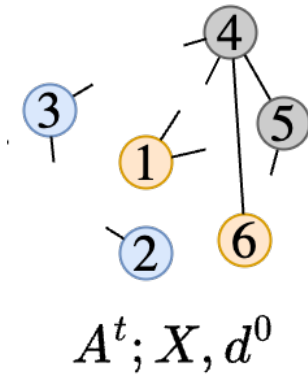
Reverse Process



Reverse Process

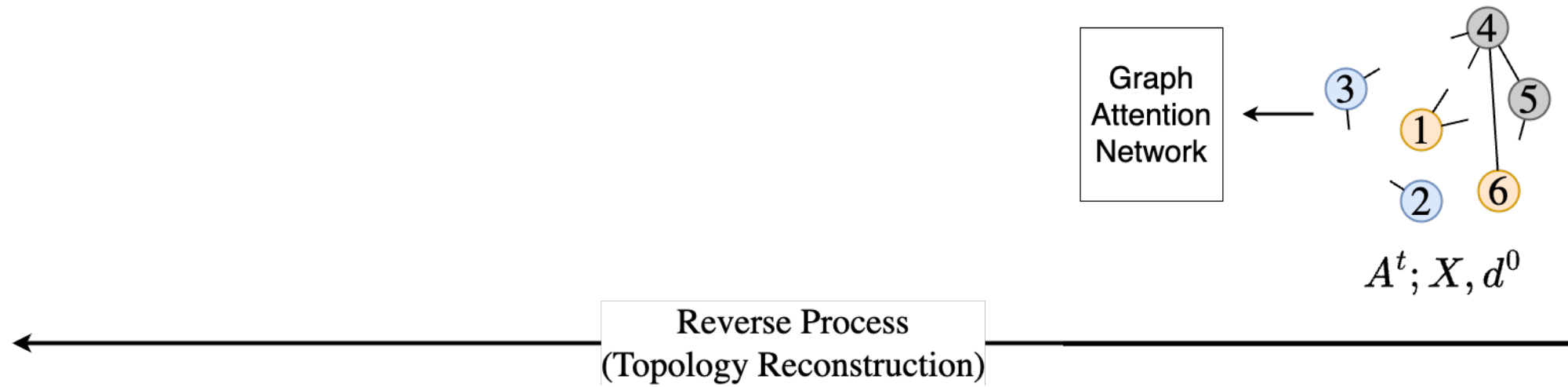


The Link Predictor

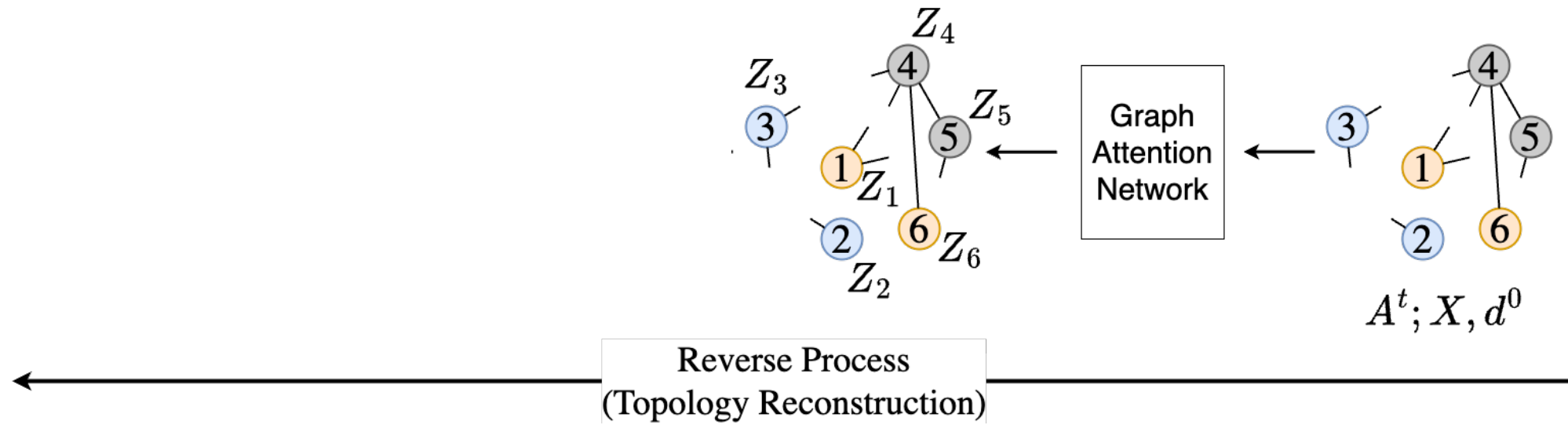


Reverse Process
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The Link Predictor

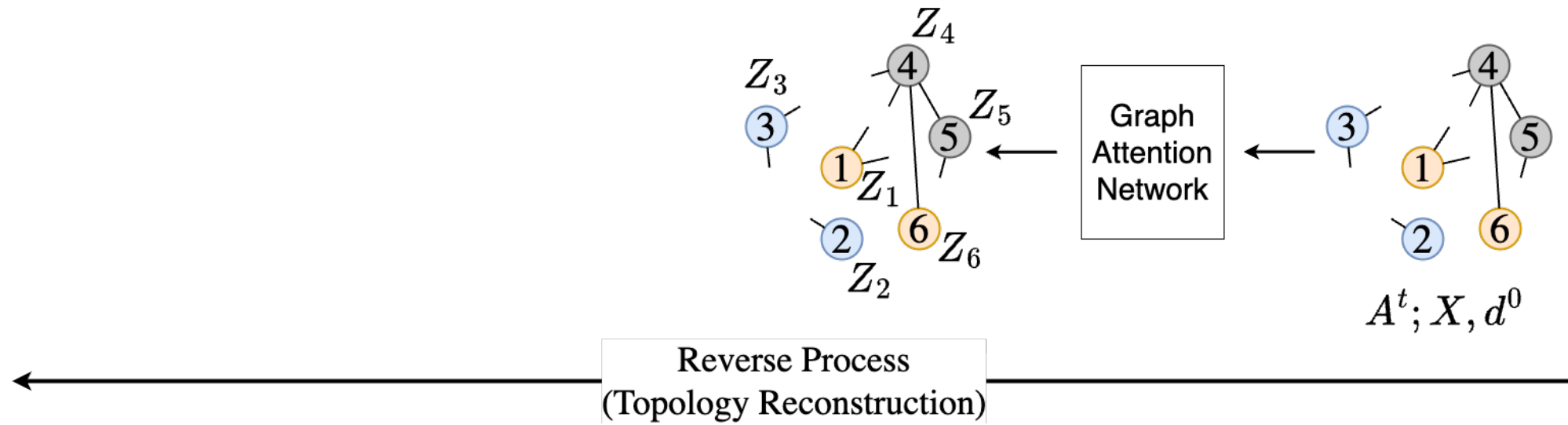


The Link Predictor

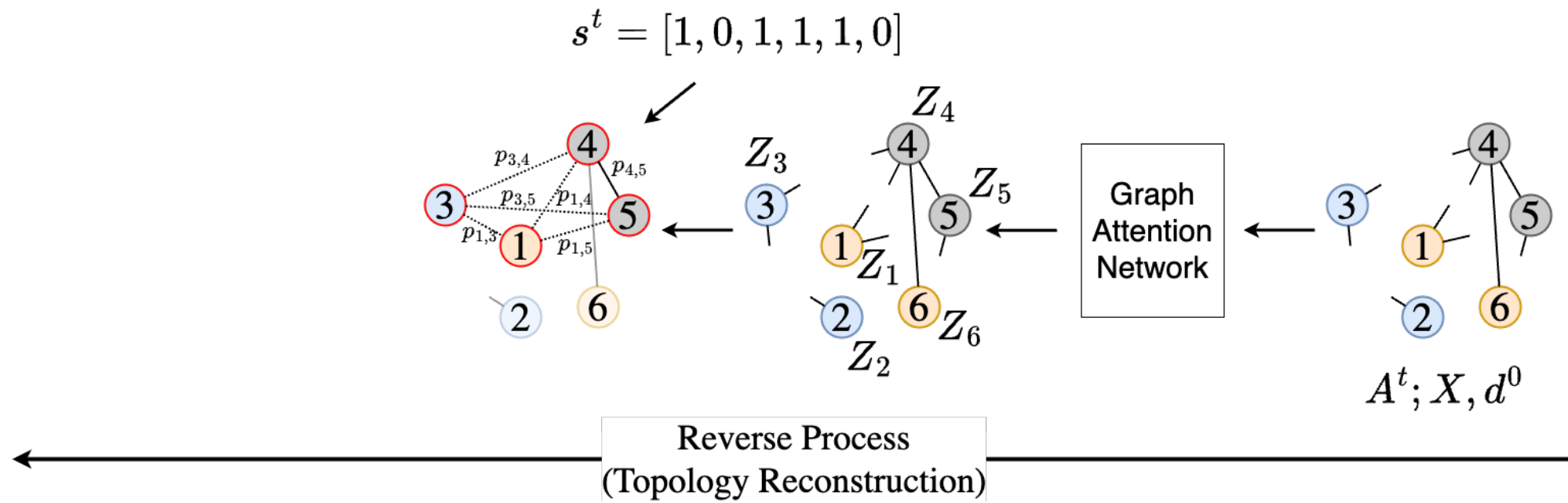


The Link Predictor

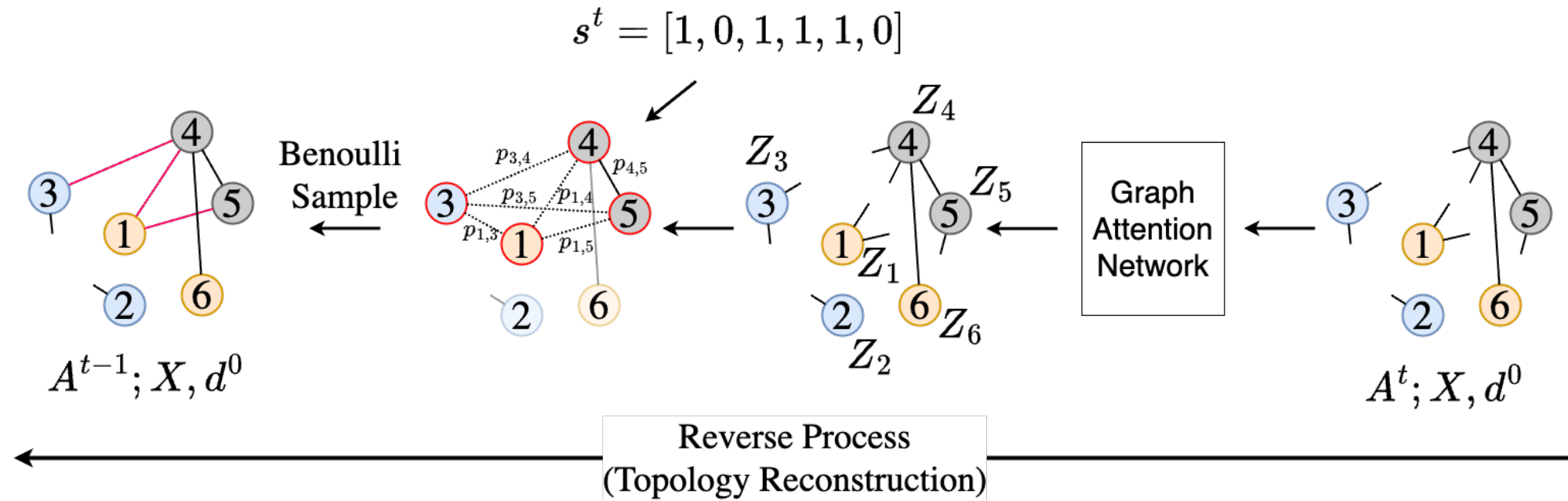
$$s^t = [1, 0, 1, 1, 1, 0]$$



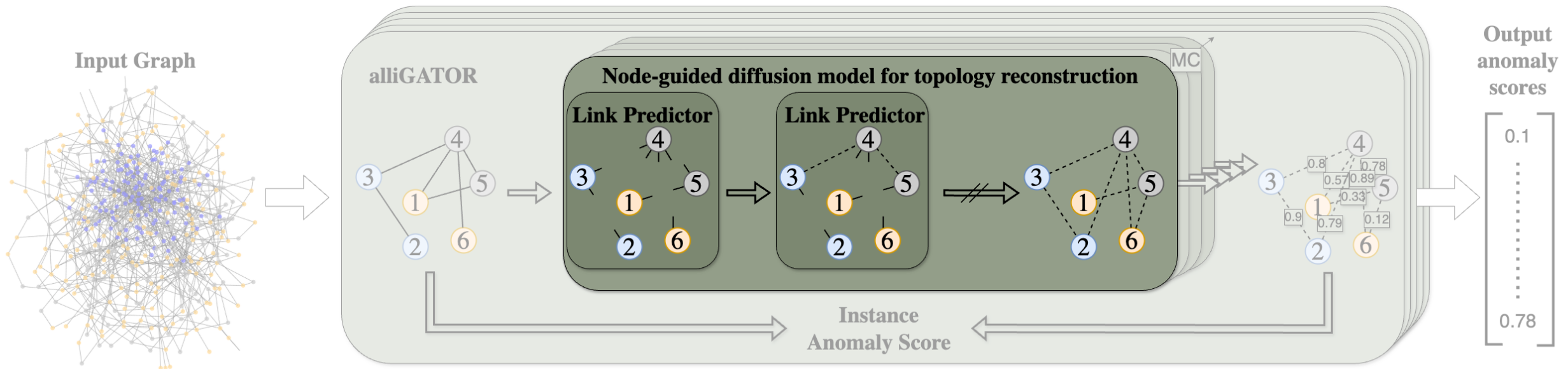
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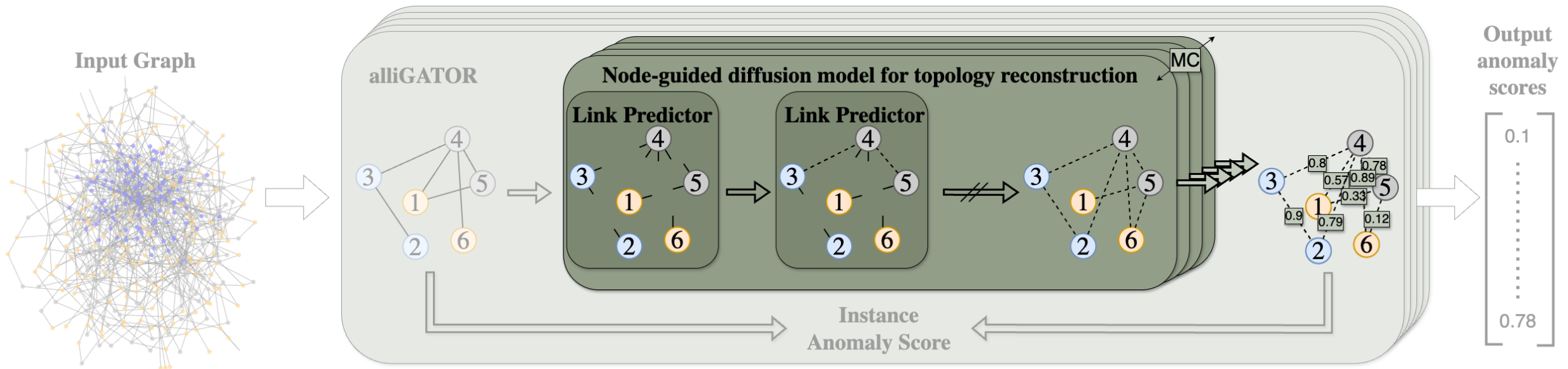
The Link Predictor



The alliGATOR model: Graph Anomaly Detection through diffusion based Topology Reconstruction

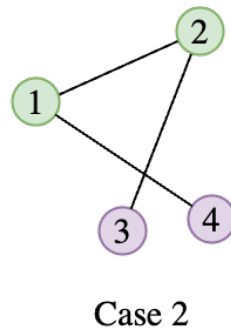
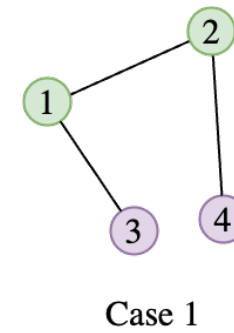
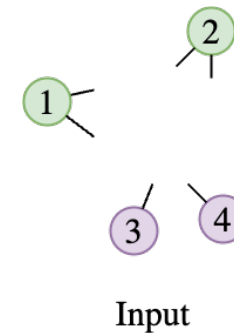


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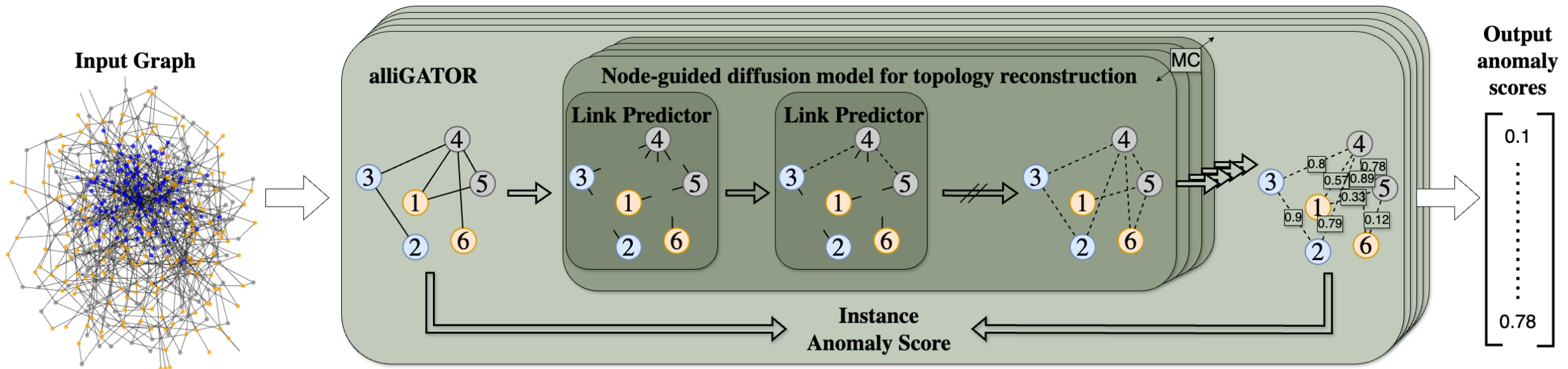


Monte Carlo Simulations

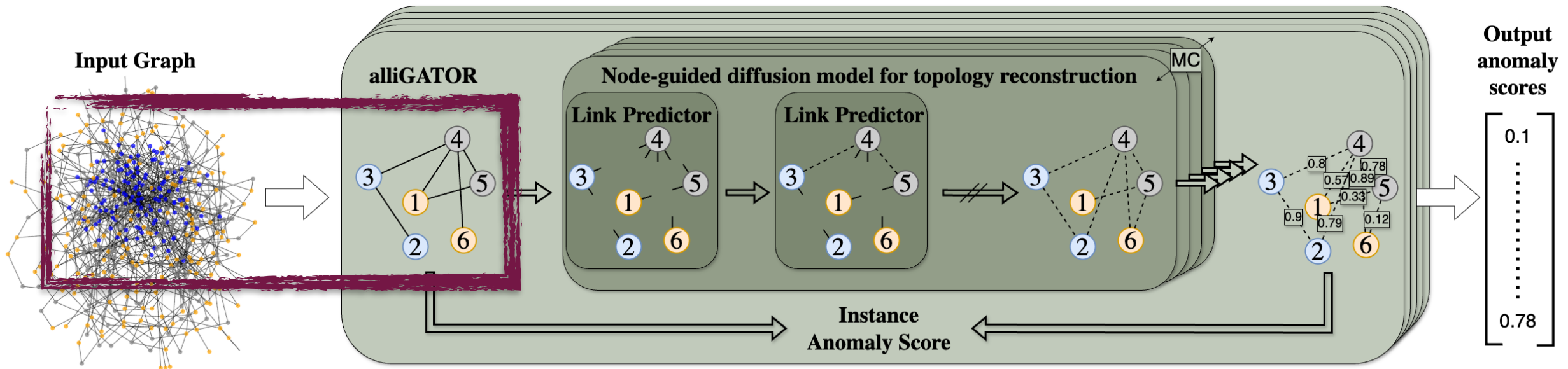
- Topology reconstructor output is definitive
(we don't get the edge probabilities)
- Multiple valid topologies
- Non-deterministic
- Obtain edge probabilities using multiple runs
- Normalise them with respect to node degrees



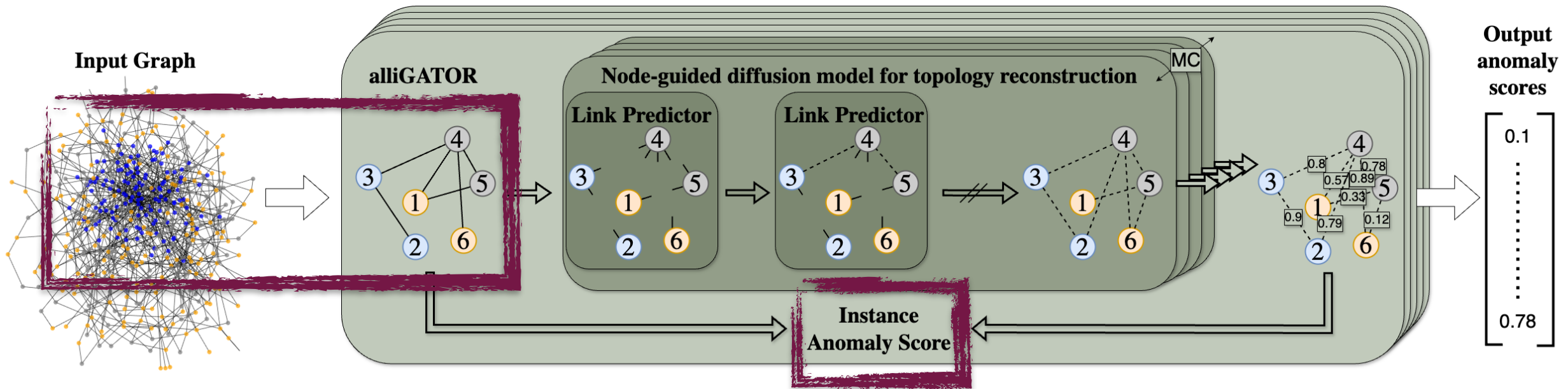
The alliGATOR model: Graph Anomaly Detection through diffusion based Topology Reconstruction



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Node Anomaly Detection

To determine the anomaly score of node c

- Personalised page rank induced subgraph of $c : G_c(N, X, E)$

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- The node attributes and degrees are passed to the topology reconstructor model $\rightarrow$ outputs the reconstructed edges with their probabilities $\rightarrow E_{rec}$

Node Anomaly Detection

To determine the anomaly score of node c

- Personalised page rank induced subgraph of $c : G_c(N, X, E)$
- The node attributes and degrees are passed to the topology reconstructor model $\rightarrow$ outputs the reconstructed edges with their probabilities $\rightarrow E_{rec}$

$$Anomaly_c = 1 - NodeActivity_c$$

$$NodeActivity_c := \mathbf{average}(\{(i, j), prob) \in E_{rec} \mid (i, j) \in E \wedge (i = c \vee j = c)\})$$

Edge Anomaly Detection

To determine the anomaly score of edge (m, n)

- Personalised page rank induced subgraph of edge $(m, n) \rightarrow G_{(m,n)}(N, X, E)$

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- The node attributes and degrees are passed to the topology reconstructor model $\rightarrow$ outputs the reconstructed edges with their probabilities $\rightarrow E_{rec}$

$$Anomaly_{(m,n)} = 1 - prob_{(m,n)}$$
$$prob(m, n) = \begin{cases} p & \text{if } ((m, n), p) \in E_{rec} \\ 0 & \text{otherwise} \end{cases}$$

Evaluation

- Dataset
- Models
- Evaluation metrics
- Results

Evaluation

- Dataset
 - Nodes are assigned to multiple classes
 - The connections between nodes are governed by the relations between the classes
 - 3 classes : blue, orange, grey

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	Blue	Orange	Grey
Blue	Barabási–Albert, 3	None	Erdős–Rényi, 0.01
Orange	None	None	Configuration Model, 7
Grey	None	Barabási–Albert, 8	None

Evaluation

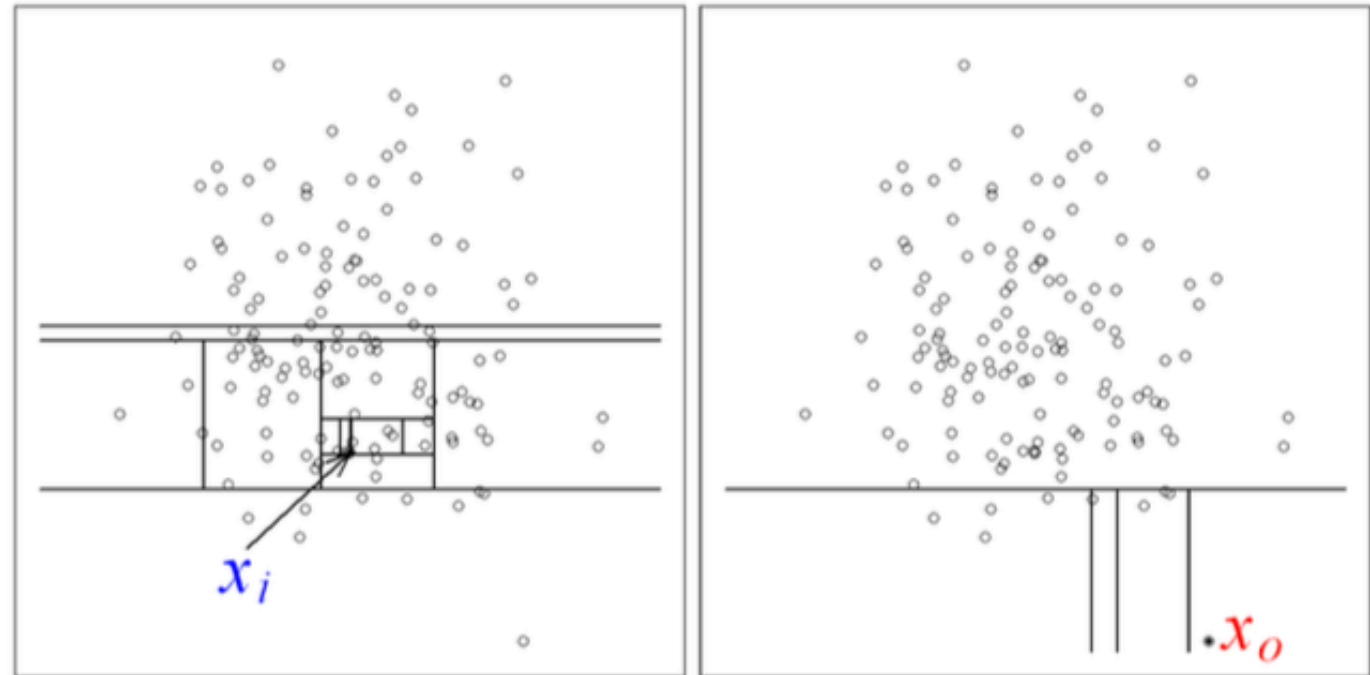
- Dataset
 - Nodes are assigned to multiple classes
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 - 3 classes : blue, orange, grey
 - ~ 1 000 nodes and ~ 7 000 edges
 - Inductive setting, splitting the edges

Evaluation

- Dataset
 - Nodes are assigned to multiple classes
 - The connections between nodes are governed by the relations between the classes
 - 3 classes : blue, orange, grey
 - ~ 1 000 nodes and ~ 7 000 edges
 - Inductive setting, splitting the edges
 - Distinct graph for each anomaly type (node and edge)
 - The anomalous instances are 4% of total

Evaluation

- Models
 - alliGATOR
 - Isolation Forest



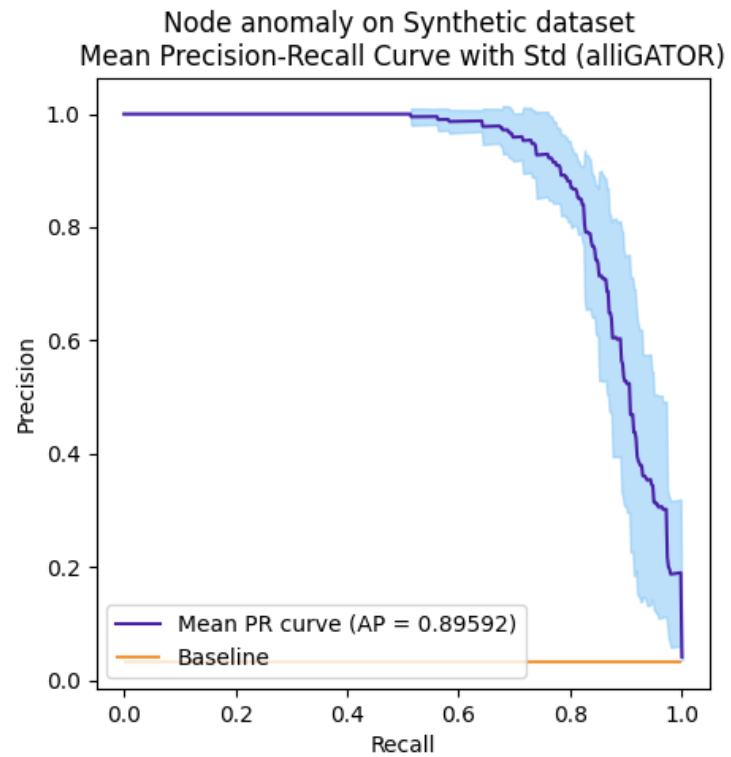
Evaluation

- Models
 - alliGATOR
 - Isolation Forest
 - Multiple data-induced random proper binary trees
 1. An attribute is selected at random
 2. A split value between the min and max of the attribute is random selected
 - Anomaly score is a function of the average path length to the instance over all trees

Evaluation

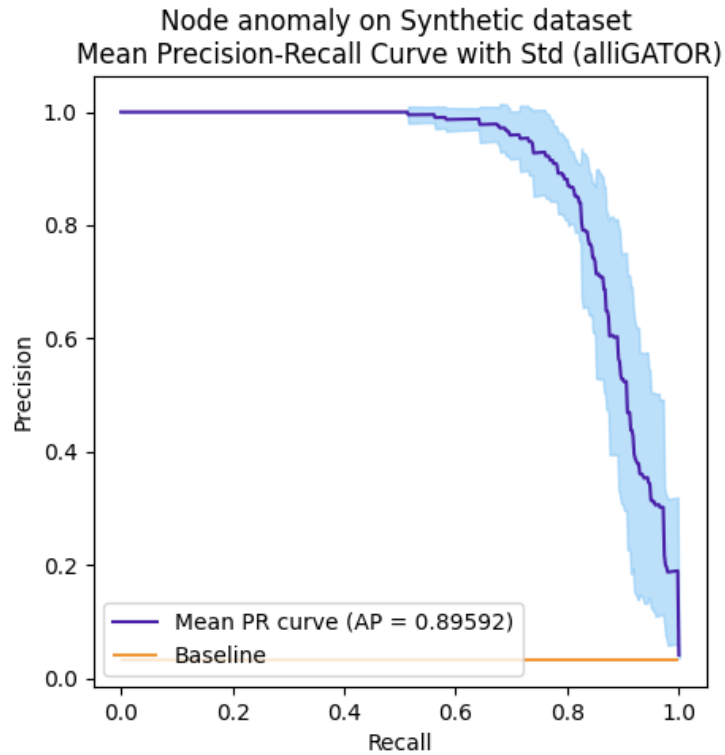
- Models
 - alliGATOR
 - Isolation Forest
- Evaluation metrics
 - Precision-recall curve
 - Average precision

Results : Node Anomaly Detection

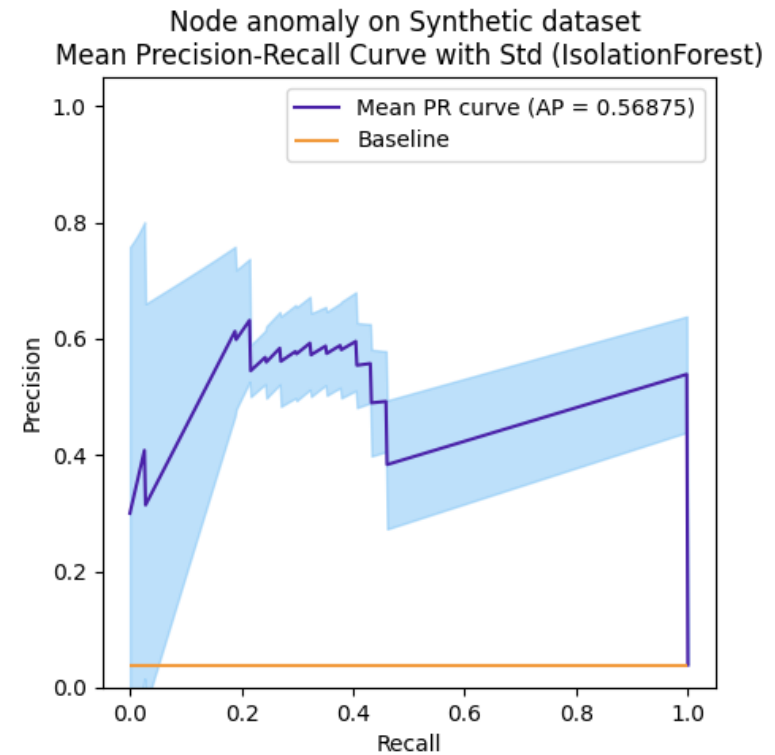


alliGATOR (ours)
AP = 0.895

Results : Node Anomaly Detection

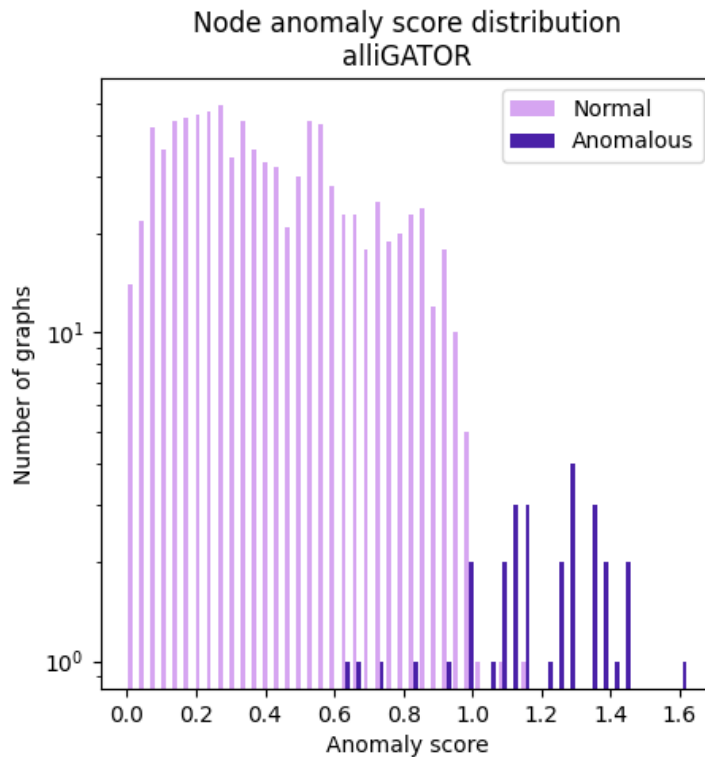


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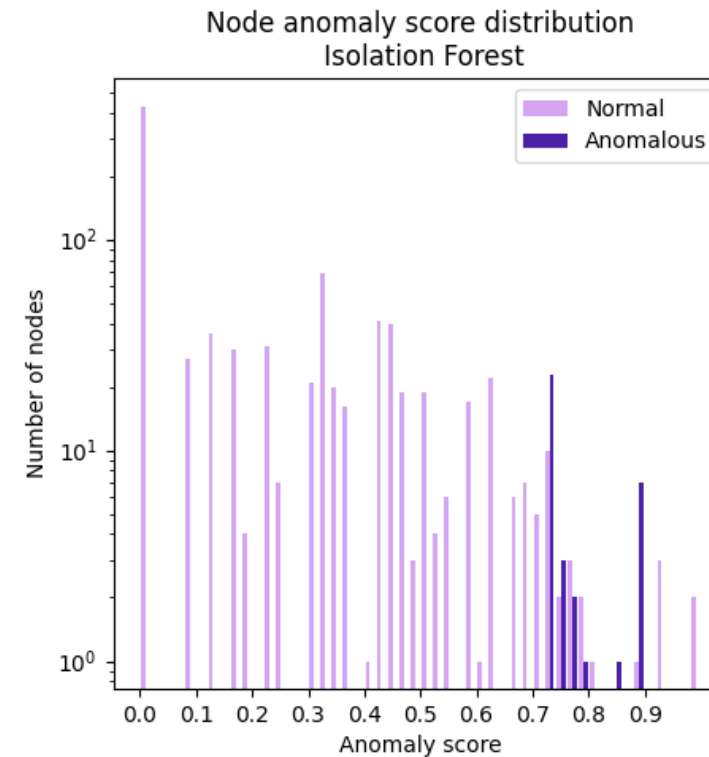


Isolation Forest
AP = 0.568

Results : Node Anomaly Detection

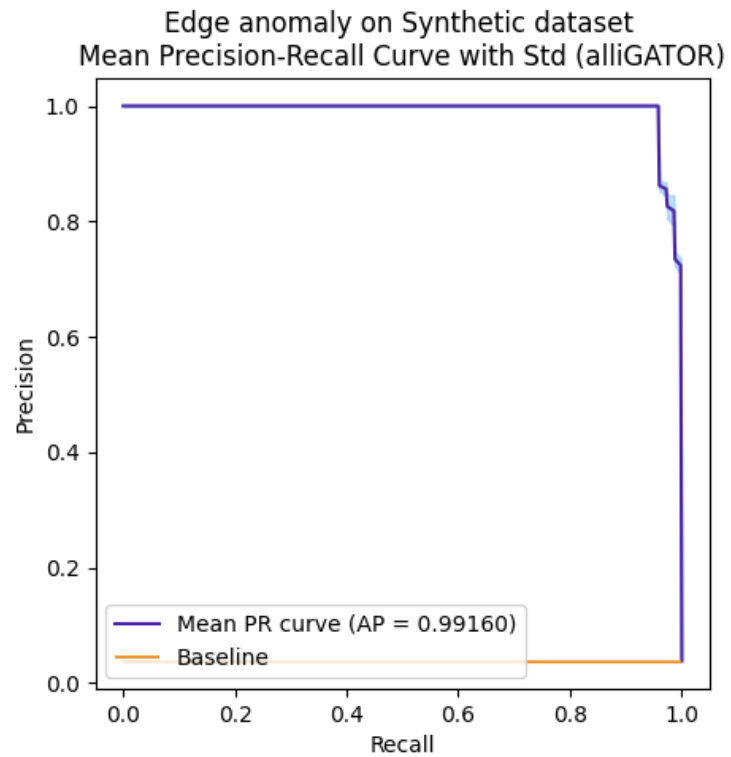


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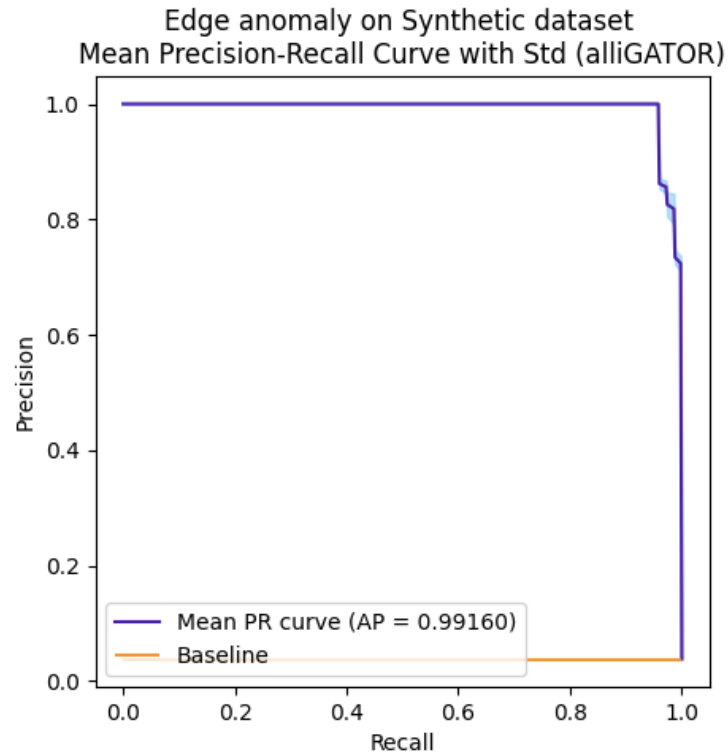
Isolation Forest
AP = 0.568

Results : Edge Anomaly Detection

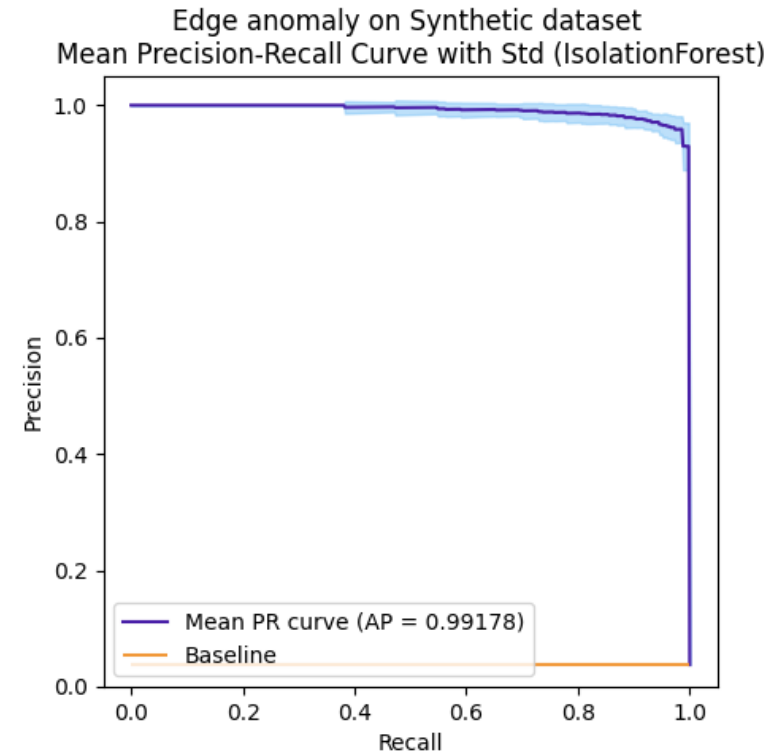


alligator (ours)
AUC = 0.99160

Results : Edge Anomaly Detection

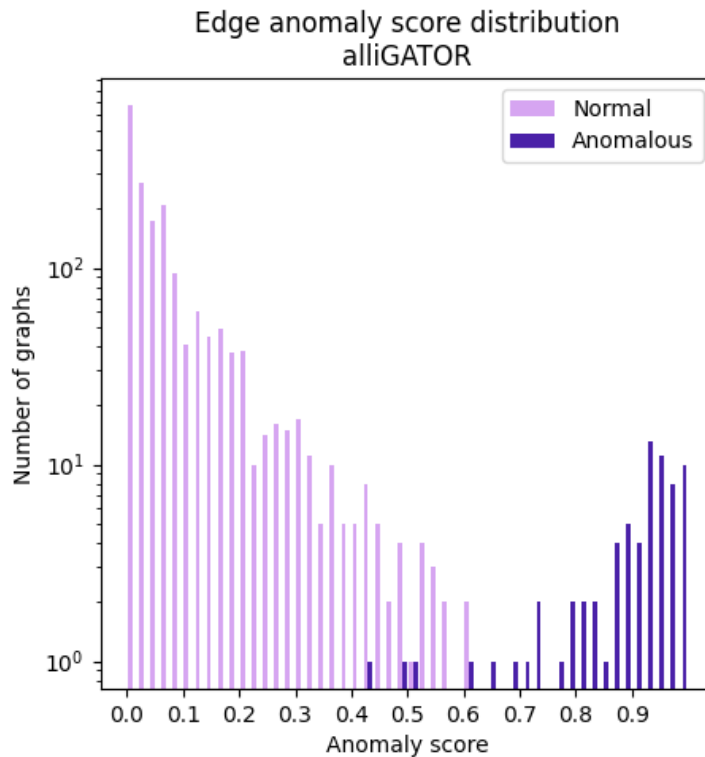


alliGATOR (ours)
AP = 0.99160

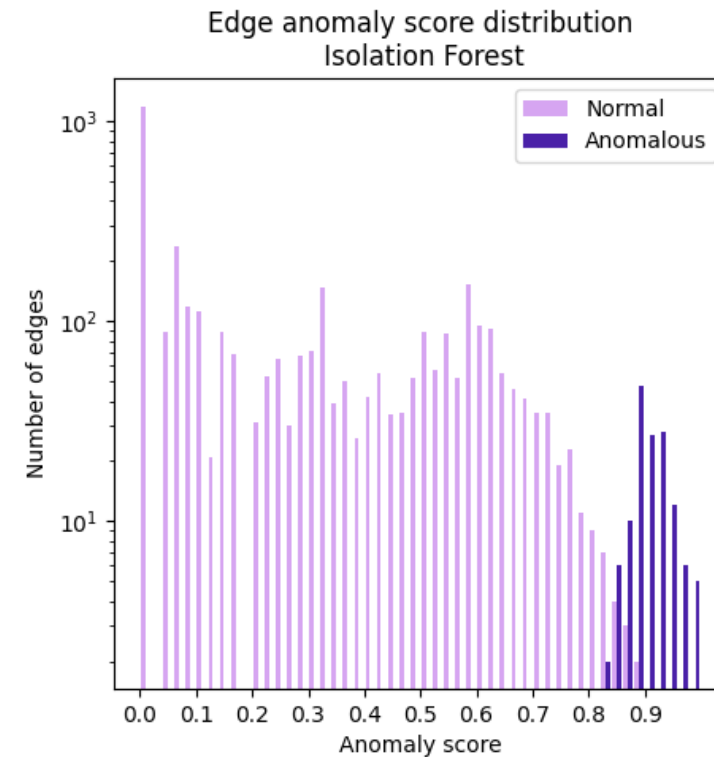


Isolation Forest
AP = 0.99178

Results : Edge Anomaly Detection



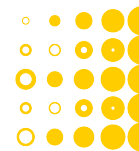
alliGATOR (ours)
AUC = 0.99160



Isolation Forest
AUC = 0.99178

Conclusions

- aliGATOR effectively detects node and edge anomalies
- Topology reconstruction can be used for graph anomaly detection
- Future work
 - Edge directionality
 - Other types of anomalies
 - Scalability to real-world datasets
 - Detecting group-based anomalies such as money laundering



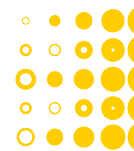


**Utrecht
University**

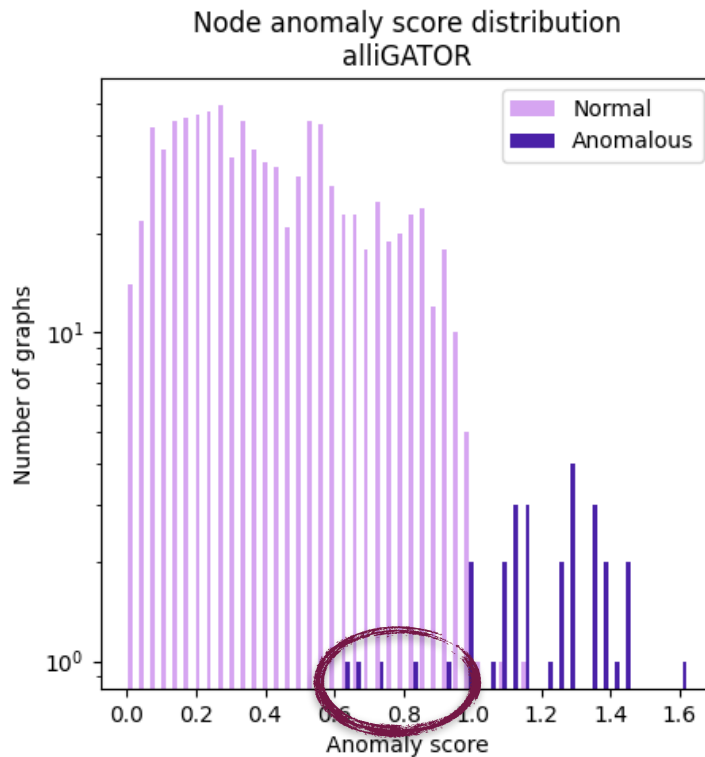
Sharing science,
shaping tomorrow

Model Limitations

- The model is sensitive to the size of the instance subgraph. If the central node is the one with the highest degree, then the model has a hard time detecting an anomaly.
 - Alternative would be to generate the topology of the whole input graph, allowing also for detecting group-based fraud
- The model is not built to detect anomalies that are purely due to degrees, i.e. high degree burst, clique anomalies, etc.



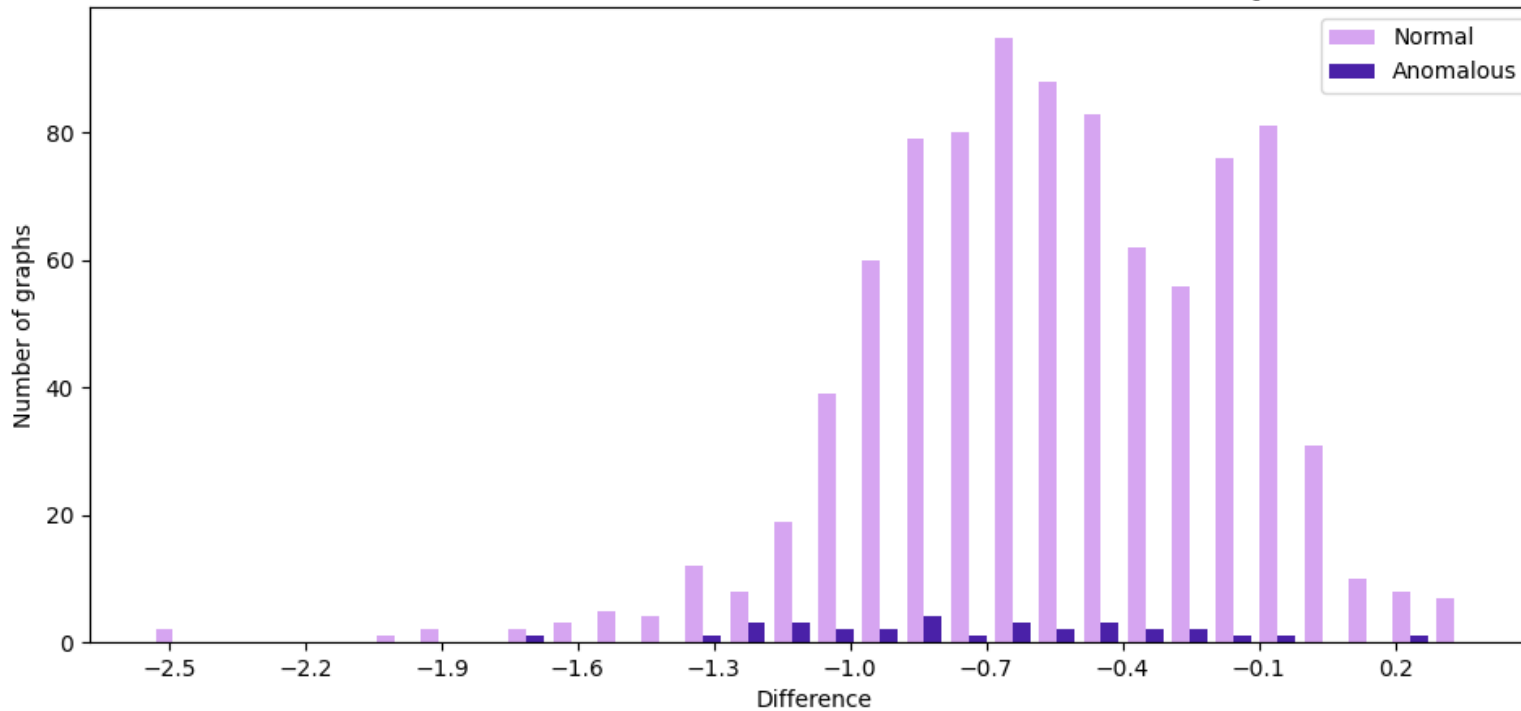
Are these really anomalous, or are they mislabelled?



- ▶ These are anomalies
- ▶ They are graphs, where the central node has high degree (7+), and the rest of the nodes have only a few edges between them.
- ▶ Solution would be to add more nodes to the instance subgraph, or to apply the reconstruction to the whole input graph

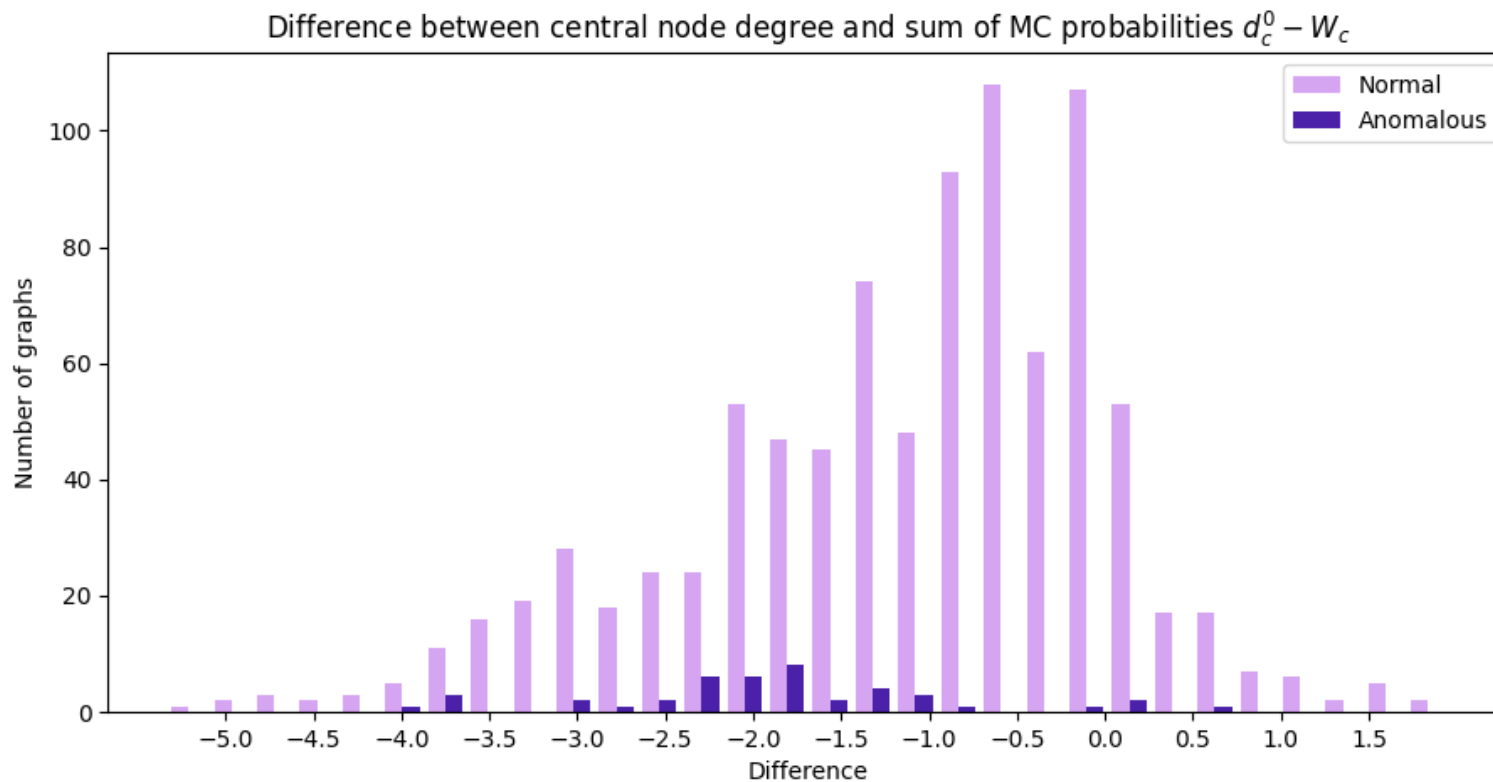
Does the model obey the degrees?

Difference between central node degree and sum of MC probabilities $d_c^0 - W_c$



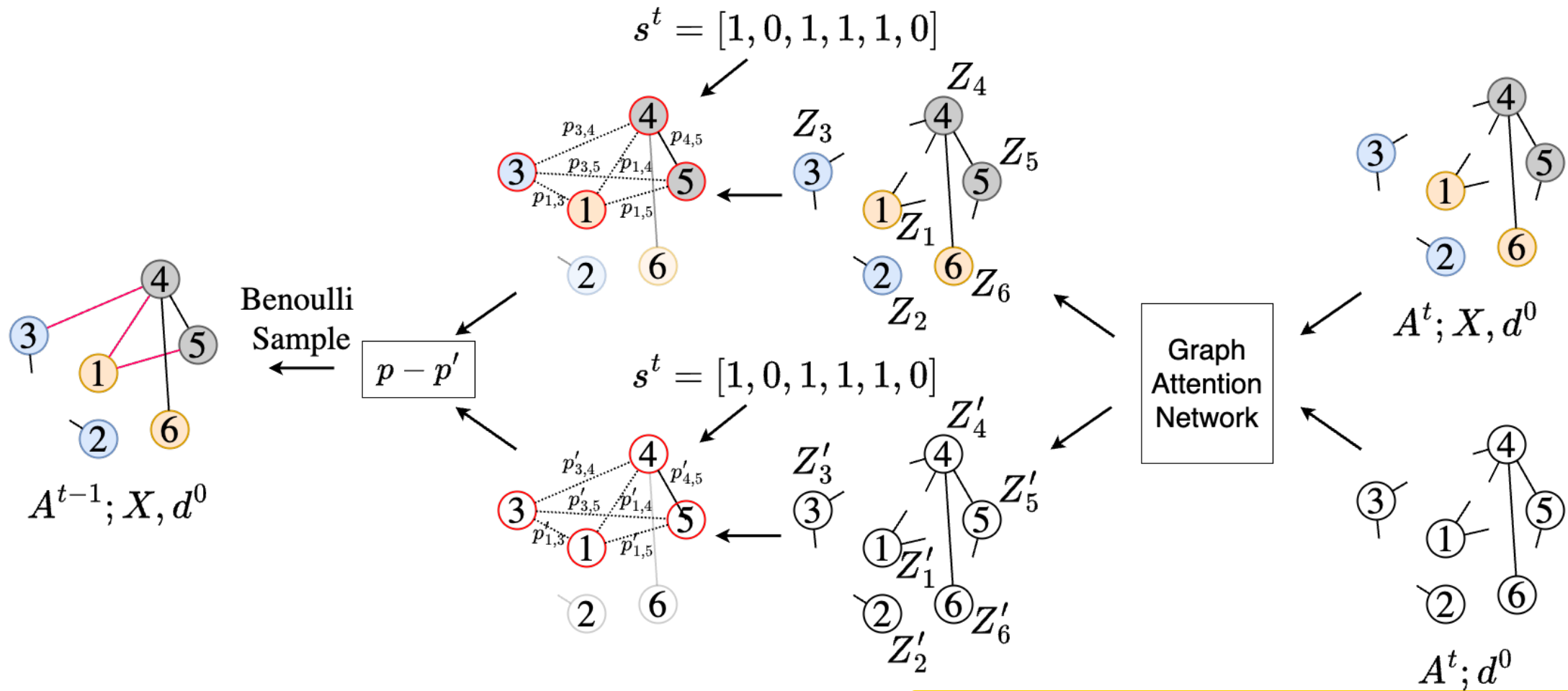
- Yes to an extent, but..

Does the model obey the degrees?



- Yes to an extent, but..
 - If we increase the guidance, the reliance on the degrees decreases
 - But with another topology reconstructor, that is not govern by degrees, it is possible to detect degree related anomalies

The link predictor with guidance



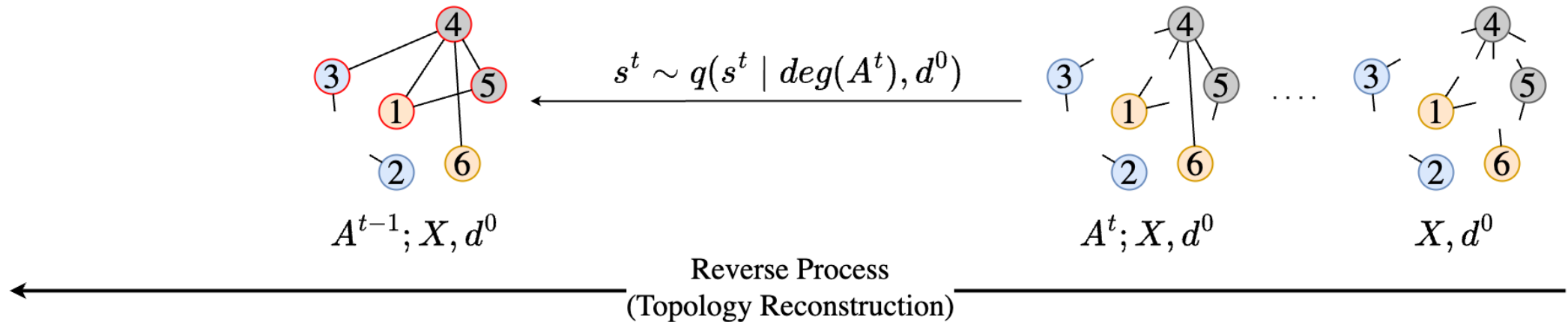
The active nodes distribution

$$q(s^t | d^t, d^0) = \prod_{i=1}^N q(s_i^t | d_i^t, d_i^0)$$

$$q(s_i^t | d_i^t, d_i^0) = \mathcal{B}(s_i^t; 1 - (1 - \gamma_t)^{\Delta_i^0})$$

$$\Delta_i^0 = d_i^0 - d_i^t$$

$$\gamma_t = \frac{\beta_t \bar{\alpha}_t - 1}{1 - \bar{\alpha}_t}$$

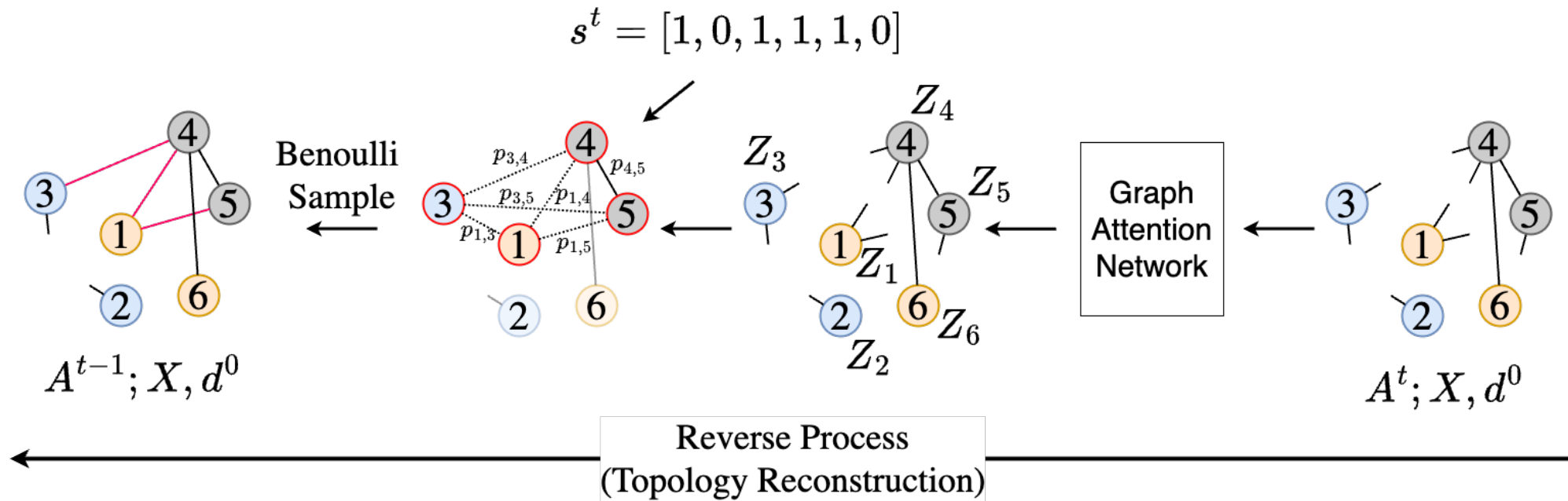


The link predictor

$$p_{\theta}(A^{t-1}|A^t, s^t, d^0, X) = \prod_{i,j;i < j} p_{\theta}(A_{i,j}^{t-1}|A_{i,j}^t, s_{i,j}^t, d^0, X)$$

$$p_{\theta}(A_{i,j}^{t-1}|A_{i,j}^t, s_{i,j}^t, d^0, X) = \mathcal{B}(A_{i,j}^{t-1} | s_{i,j}^t \mu_{\theta}(A^t, s^t, d^0, X, t) + (1 - s_{i,j}^t) A_{i,j}^t)$$

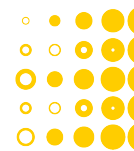
$$s_{ij}^t = s_i^t s_j^t$$



The Topology Reconstructor Loss

$$\mathcal{L}(A^0|d^0, X; \theta) = \mathbb{E}_q \left[\log \frac{p(A^T)}{q(A^T|A^0)} + \underbrace{\log p_\theta(A^0|A^1, s^1, d^0, X)}_{\text{reconstruction term } \mathcal{L}_{rec}} + \sum_{t=2}^T \underbrace{\log \frac{p_\theta(A^{t-1}|A^t, s^t, d^0, X)}{q(A^{t-1}|A^t, s^t, A^0)}}_{\text{edge prediction term } \mathcal{L}_{edge}(t)} \right] \quad (3.2)$$

$$\nabla_\theta \mathcal{L}(A^0|d^0, X; \theta) = \mathbb{E}_{t \sim \mathcal{N}[1, T]} \left[\mathbb{E}_{q(A^{t-1}|A^0)} \mathbb{E}_{q(A^t|A^{t-1})} \mathbb{E}_{q(s^t|A^{t-1}, A^t)} [\log p_\theta(A^{t-1}|A^t, s^t, d^0, X)] \right] \quad (3.3)$$



Model Implementation

$$\mathbf{Z}^0 = \text{emb}(\mathbf{d}^t) \parallel \text{emb}(\mathbf{d}^0) \parallel \text{emb}(\mathbf{s}^t) \parallel \text{emb}(\mathbf{X})$$

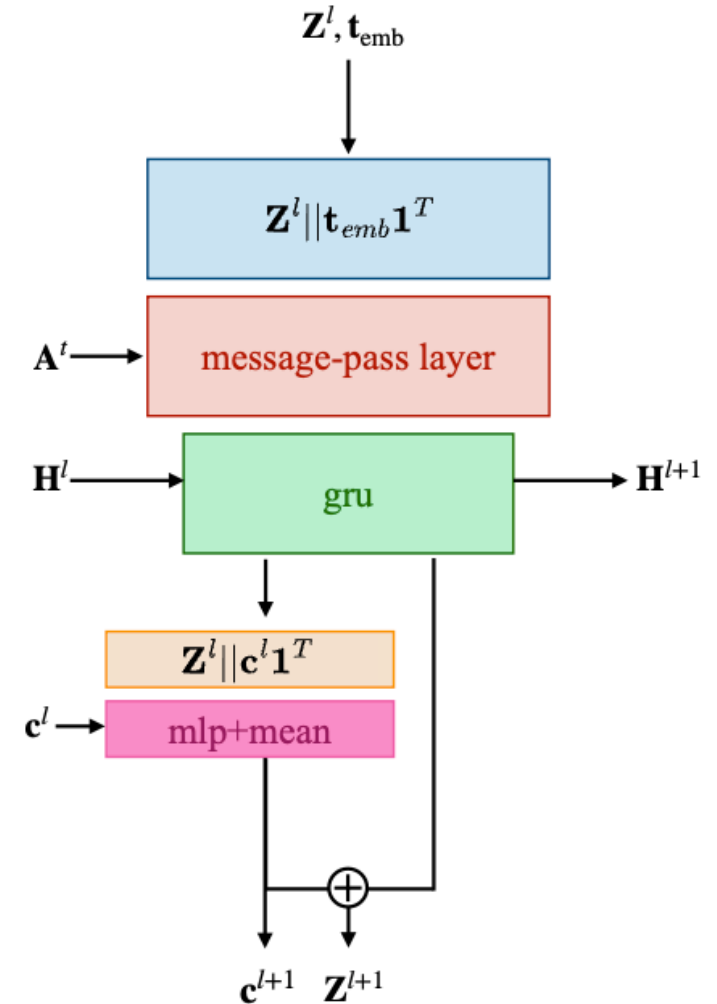
$$t_{\text{emb}} = \text{emb}(t)$$

$$\mathbf{c}^0 = \text{mean}(\mathbf{Z}^0)$$

$$\mathbf{H}^0 = \mathbf{Z}^0$$

$$\mathbf{Z}^l, \mathbf{c}^l, \mathbf{H}^l = \text{block}_l(\mathbf{Z}^{l-1}, t_{\text{emb}}, \mathbf{c}^{l-1}, \mathbf{H}^{l-1}), \text{ for } l = 1, \dots, L$$

$$p_{(i,j)}^{t-1} = \text{mlp}(\mathbf{Z}_i^L, \mathbf{Z}_j^L) \text{ for } (i,j) \in \{(i,j) \mid s_{ij}^t = 1, 1 \leq i < j \leq V\}$$



alliGATOR Complexity

- Personalised PageRank has complexity of $O(n)$
- Topology reconstructor
 - $O(n' \cdot f \cdot f' + m' \cdot f')$ per layer
 - $O(n')$ predict the active nodes
 - $O(k^2)$ predict the links between k active nodes
 - $O(T \cdot \max(k^2, m))$ for the diffusion process
 - Monte Carlo if parallelised doesn't change it

How do we normalise the edge probabilities with respect to the node degrees?

The expected number of edges between two nodes is $\frac{k_i k_j}{2m}$

Additional probability/score $L_{i,j} = W_{i,j} - \frac{k_i k_j}{2m}$

- >1 model is confident the edge should exist
- <1 model is confident the edge should not exist

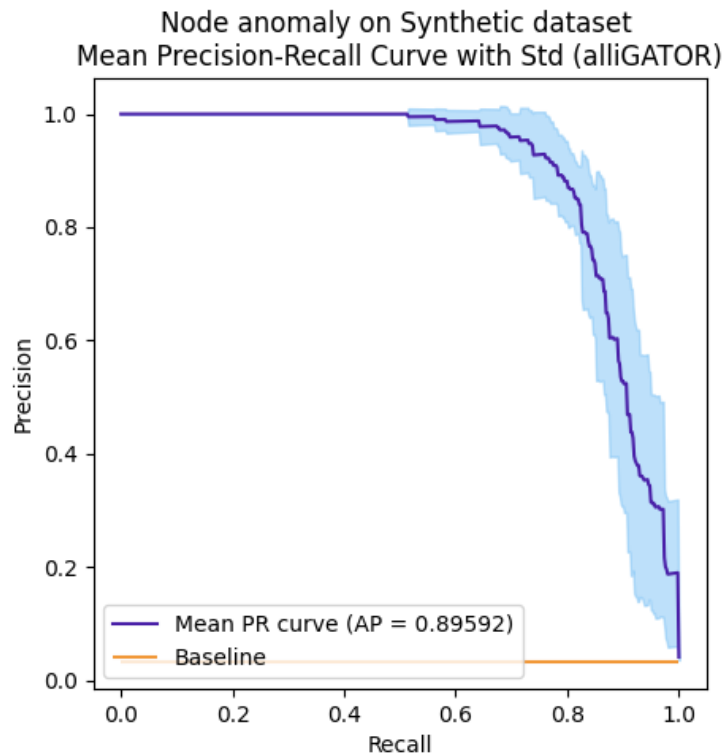
Node anomaly the node activity is never 1, so we want to normalise it, therefore :

$$S_c = \frac{\frac{1}{|\mathcal{N}(c)|} \sum_{j \in \mathcal{N}(c)} L_{c,j}}{1 - \frac{1}{2m} \sum_{j \in \mathcal{N}(i)} k_j}$$

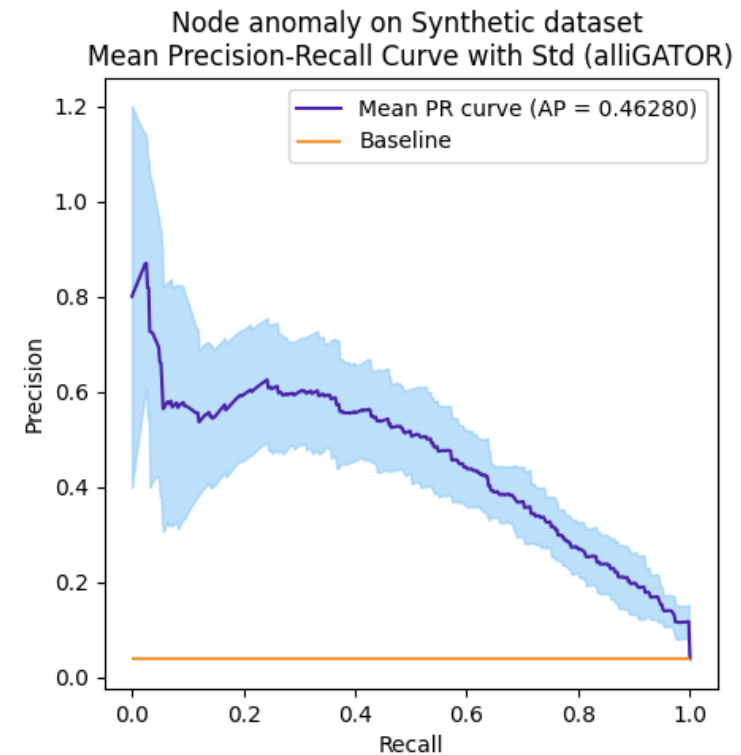
The node activity score measure the level of agreement between the original and the reconstructed topology it is upper bounded by 1, but can be less than 0.

The anomaly score is the complement of the score.

Edge Normalisation Effect



alliGATOR (ours)
AP = 0.895



alliGATOR without edge normalisation
AP = 0.462

Diffusion models for graph generation

- Applying the original continuous diffusion
 - on the adjacency matrix (Niu et al, 2020, Huang et al, 2022)
 - on both : node feature matrix & the adjacency matrix (Jo et al, 2022)
 - *limitation*: applying continuous noise on discrete matrix
- Discrete diffusion models
 - noise from discrete distribution instead of continuous
Haefeli et al. (2022), Vignac et al. (2022), Chen et al. (2023)

Anomaly score function of Isolation Forest (1/2)

- $h(x)$ is a path length
- $E(h(x))$ is the average path over the trees
 - We want to normalise the scores of all instances
 - The maximum height of tree grows linearly with the number of instances in the dataset (n)
 - The average height grows in order of $\log n$
 - Binary search trees have equivalent structures, the average path length of a tree is

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}$$

H is Euler's constant

Anomaly score function of Isolation Forest (2/2)

- $h(x)$ is a path length
- $E(h(x))$ is the average path over the trees
 - The average path length of a tree is

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}$$

- The anomaly score of instance x is

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

Isolation Forest Complexity

- t - number of trees
- ψ - number of subsamples
- n - number of evaluating instances
- During training $O(t \cdot \psi \cdot \log \psi)$
- During evaluation $O(n \cdot t \cdot \log \psi)$